

SIVKORT STERILE SUSPENSION FOR INJECTION 40MG/1ML  
Triamcinolone Acetonide 40MG/1ML sterile suspension for injection

**Product Name**

SIVKORT STERILE SUSPENSION FOR INJECTION 40MG/1ML

**Product Description**

A white suspension liquid in amber glass ampoules.

**Composition**

Each 1 ml injection ampoule contains 40 mg of Triamcinolone Acetonide and 10 mg of Benzyl Alcohol.

**Pharmacodynamics**

Triamcinolone Acetonide is a glucocorticoid steroid. Its metabolism of action is the same as other glucocorticoid steroids.

**Pharmacokinetics**

Triamcinolone acetonide is a synthetic glucocorticoid with marked anti-inflammatory and anti-allergic actions.

Intramuscular Injection: Provides an extended duration of therapeutic effect and fewer side effects of the kind associated with oral corticosteroid therapy, particularly gastro-intestinal reactions such as peptic ulceration. Studies indicate that, following a single intramuscular dose of 80mg triamcinolone acetonide, adrenal suppression occurs within 24 - 48 hours and then gradually returns to normal, usually in approximately three weeks. This finding correlates closely with the extended duration of therapeutic action of triamcinolone acetonide.

Triamcinolone acetonide may be absorbed into the systemic circulation from synovial spaces. Triamcinolone acetonide is absorbed slowly, though almost completely, following depot administration by deep intramuscular injection; biologically active levels are achieved systemically for prolonged periods (weeks to months). In common with other corticosteroids, triamcinolone is metabolised largely hepatically but also by the kidney and is excreted in urine. The main metabolic route is 6-beta-hydroxylation; no significant hydrolytic cleavage of the acetonide occurs.

In view of the hepatic metabolism and renal excretion of triamcinolone acetonide, functional impairments of the liver or kidney may affect the pharmacokinetics of the drug.

**Indication**

Intramuscular

Where oral therapy is not feasible or is temporarily undesirable in the judgment of the physician, SIVKORT® INJECTION is indicated for intramuscular use as follows:

1. Endocrine disorders  
Nonsuppurative thyroiditis.

## 2. Rheumatic disorders

As adjunctive therapy for short-term administration in: post-traumatic osteoarthritis; synovitis of osteoarthritis; rheumatoid arthritis; acute and subacute bursitis; epicondylitis; acute non-specific tenosynovitis; acute gouty arthritis; psoriatic arthritis; ankylosing spondylitis; juvenile rheumatoid arthritis.

## 3. Collagen diseases

During an exacerbation or as maintenance therapy in selected cases of systemic lupus erythematosus: acute rheumatic carditis.

## 4. Dermatologic diseases

Pemphigus; severe erythema multiforme (Stevens-Johnson syndrome); exfoliative dermatitis; bullous dermatitis herpetiformis; severe seborrheic dermatitis; severe psoriasis.

## 5. Allergic states

Control of severe or incapacitating allergic conditions intractable to adequate trials of conventional treatment in bronchial asthma; contact dermatitis; atopic dermatitis; seasonal or perennial allergic rhinitis.

## 6. Ophthalmic diseases

Severe chronic allergic, and inflammatory processes involving the eye, such as: herpes zoster ophthalmicus; iritis; iridocyclitis; chorioretinitis; diffuse posterior uveitis and choroiditis; optic neuritis; sympathetic ophthalmia; anterior segment inflammation.

## 7. Gastrointestinal diseases

Critical phase requiring systemic therapy in ulcerative colitis, regional enteritis.

## 8. Respiratory diseases

Symptomatic sarcoidosis; berylliosis; aspiration pneumonitis.

## 9. Hematologic diseases

Acquired (autoimmune) hemolytic anemia.

## 10. Neoplastic disorders

For palliative management of: leukemias and lymphomas in adults; acute leukemia of childhood.

## 11. Edema

To induce diuresis or remission of proteinuria in the nephrotic syndrome, without uremia, of the idiopathic type or that due to lupus erythematosus.

Intra-Articular SIVKORT® INJECTION is indicated for intra-articular or intra bursal administration, and for injections into tendon sheaths, as adjunctive therapy for short-term administration in the following conditions: synovitis of osteoarthritis; rheumatoid arthritis; acute and subacute bursitis; acute gouty arthritis; epicondylitis; acute nonspecific tenosynovitis; posttraumatic osteoarthritis.

### **Recommended Dosage**

General This preparation contains benzyl alcohol. Not for use in newborn or premature infants (see USE IN CHILDREN).

The initial dose of SIVKORT® INJECTION may vary from 2.5 to 60 mg per day depending on the specific disease entity being treated (see Dosage section below). In less severe conditions, lower doses generally suffice, while in other patients, higher initial doses may be required. Usually the parenteral dosage range is one-third to one-half the oral dose, given every 12 hours. In life-threatening situations, administration of higher dosages may be justified. The initial dosage should be maintained or adjusted until a satisfactory response is noted. If after a reasonable period of time there is a lack of satisfactory clinical response, SIVKORT® INJECTION should be gradually discontinued and the patient transferred to other appropriate therapy.

DOSAGE REQUIREMENTS ARE VARIABLE AND MUST BE INDIVIDUALIZED ON THE BASIS OF THE DISEASE UNDER TREATMENT AND THE RESPONSE OF THE PATIENT. The lowest possible dose of corticosteroid should be used to control the condition being treated. After a favorable response is noted, the proper maintenance dosage should be determined by decreasing the initial drug dosage in small decrements at appropriate time intervals down to the lowest dosage which will maintain the desired clinical response. Constant monitoring of drug dosage is necessary. Dose adjustments may be necessary in accordance with changes in clinical status. Patient exposure to stressful situations not directly related to the disease may necessitate increasing the dosage. After long-term therapy, it is recommended that SIVKORT® INJECTION be withdrawn gradually.

#### Dosage

##### Systemic

Adults and children over 12 years of age: The suggested initial dose is 60 mg, injected deeply into the gluteal muscle. Atrophy of subcutaneous fat may occur if the injection is not properly given. Dosage is usually adjusted within the range of 40 to 80 mg, depending upon patient response and duration of relief. However, some patients may be well controlled on doses as low as 20 mg or less.

Hay fever or pollen asthma: Patients with hay fever or pollen asthma who are not responding to pollen administration and other conventional therapy may obtain a remission of symptoms lasting throughout the pollen season after a single injection of 40 to 100 mg.

Children 6 to 12 years: The suggested initial dose is 40 mg although dosage depends more on the severity of symptoms than on age or weight.

##### Local

Intra-articular or intrabursal administration and for injection into tendon sheaths: A single local injection of triamcinolone acetonide is frequently sufficient, but several injections may be needed for adequate relief of symptoms.

Initial dose: 2.5 to 5 mg for smaller joints and from 5 to 15 mg for larger joints, depending on the specific disease entity being treated. For adults, doses up to 10 mg for smaller areas and up to 40 mg for larger areas have usually been sufficient. Single injections into several joints, up to a total of 80 mg, have been given without undue reactions.

## Administration

### General

STRICT ASEPTIC TECHNIQUE IS MANDATORY. The vial should be shaken before use to ensure a uniform suspension. Prior to withdrawal, the suspension should be inspected for clumping or granular appearance (agglomeration). An agglomerated product results from exposure to freezing temperatures and should not be used. After withdrawal, SIVKORT® INJECTION should be injected without delay to prevent setting in the syringe. Careful technique should be employed to avoid the possibility of entering a blood vessel or introducing infection.

### Systemic

For systemic therapy, injection should be made deeply into the gluteal muscle (see WARNINGS). For adults, a minimum needle length of 1½ inches is recommended. In obese patients, a longer needle may be required. Use alternative sites for subsequent injections.

### Local

For treatment of joints, usual intra-articular injection techniques should be followed. If an excessive amount of synovial fluid is present in the joint, some, but not all, should be aspirated to aid in the relief of pain and to prevent undue dilution of the steroid. With intra-articular or intrabursal administration and with injection into tendon sheaths, prior use of a local anesthetic may often be desirable. Care should be taken with this kind of injection, particularly in the deltoid region, and with injection into tendon sheaths to avoid injecting the suspension into the tissues surrounding the site since this may lead to tissue atrophy. In treating acute nonspecific tenosynovitis, care should be taken to ensure that the injection of the corticosteroid is made into the tendon sheath rather than the tendon substance. Epicondylitis may be treated by infiltrating the preparation into the area of greatest tenderness.

### **Incompatibility**

Triamcinolone Acetonide should not be combined with other aqueous solutions of medicaments except with water for injection. The injection should be freshly prepared and unused portions should be discarded.

As this preparation contains benzyl alcohol, its use should be avoided in children under two years of age. Not to be used in neonates.

### **Contraindications**

Hypersensitivity to any of the ingredients.

Systemic infections unless specific anti-infective therapy is employed.

Administration by intravenous, intrathecal or intraocular injection.

### **Warnings and Precautions**

#### Warnings

Adequate studies to demonstrate the safety of Sivkort injection use by intra-tubinal, subconjunctival, sub-tenons, retrobulbar and intraocular (intravitreal) injections have not been performed. Endophthalmitis, eye inflammation, increased intraocular pressure and visual disturbances including vision loss have been reported with intravitreal administration. Several

instances of blindness have been reported following injection of corticosteroid suspensions into the nasal turbinates and intralesional injection about the head.

**(Intra-Articular Injection):**

Corticosteroids should not be injected into unstable joints.

Patients should be specifically warned to avoid over-use of joints in which symptomatic benefit has been obtained. Severe joint destruction with necrosis of bone may occur if repeated intra-articular injections are given over a long period of time. Care should be taken if injections are given into tendon sheaths to avoid injection into the tendon itself. Repeated injection into inflamed tendons should be avoided as it has been shown to cause tendon rupture.

Due to the absence of a true tendon sheath, the Achilles tendon should not be injected with depot corticosteroids.

**(Intramuscular Injection):**

During prolonged therapy a liberal protein intake is essential to counteract the tendency to gradual weight loss sometimes associated with negative nitrogen balance and wasting of skeletal muscle.

**Precautions:**

Intra-articular injection should not be carried out in the presence of active infection in or near joints. The preparation should not be used to alleviate joint pain arising from infectious states such as gonococcal or tubercular arthritis.

Undesirable effects may be minimised using the lowest effective dose for the minimum period, and by administering the daily requirement, whenever possible, as a single morning dose on alternate days. Frequent patient review is required to titrate the dose appropriately against disease activity.

Adrenal cortical atrophy develops during prolonged therapy and may persist for years after stopping treatment. Withdrawal of corticosteroids after prolonged therapy must, therefore, always be gradual to avoid acute adrenal insufficiency and should be tapered off over weeks or months according to the dose and duration of treatment. During prolonged therapy any intercurrent illness, trauma or surgical procedure will require a temporary increase in dosage. If corticosteroids have been stopped following prolonged therapy they may need to be reintroduced temporarily.

Patients should carry steroid treatment cards which give clear guidance on the precautions to be taken to minimise risk and which provide details of prescriber, drug, dosage and the duration of treatment.

Suppression of the inflammatory response and immune function increases the susceptibility to infections and their severity. The clinical presentation may often be atypical and serious infections such as septicaemia and tuberculosis may be masked and may reach an advanced stage before being recognised.

Chickenpox and measles are of particular concern since these normally minor illnesses may be fatal in immunosuppressed patients.

Unless they have had chickenpox, patients receiving parenteral corticosteroids for purposes other than replacement should be regarded as being at risk of severe chickenpox. Manifestations of fulminant illness include pneumonia, hepatitis and disseminated intravascular coagulation; rash is not necessarily a prominent feature. Passive immunisation with varicella zoster

immunoglobulin (VZIG) is needed by exposed non-immune patients who are receiving systemic corticosteroids or who have used them within the previous 3 months; varicella-zoster immunoglobulin should preferably be given within 3 days of exposure and not later than 10 days. Confirmed chickenpox warrants specialist care and urgent treatment. Corticosteroids should not be stopped and the dose may need to be increased.

Patients should be advised to avoid exposure to measles and to seek medical advice without delay if exposure occurs. Prophylaxis with normal immunoglobulin may be needed.

During corticosteroid therapy antibody response will be reduced and therefore affect the patient's response to vaccines. Live vaccines should not be administered.

#### Special Precautions:

Particular care is required when considering use of systemic corticosteroids in patients with the following conditions and frequent patient monitoring is necessary.

- Recent intestinal anastomoses, diverticulitis, thrombophlebitis, existing or previous history of severe affective disorders (especially previous steroid psychosis), exanthematous disease, chronic nephritis, or renal insufficiency, metastatic carcinoma, osteoporosis (post-menopausal females are particularly at risk); in patients with an active peptic ulcer (or a history of peptic ulcer). Myasthenia gravis. Latent or healed tuberculosis; in the presence of local or systemic viral infection, systemic fungal infections or in active infections not controlled by antibiotics. In acute psychoses; in acute glomerulonephritis. Hypertension; congestive heart failure; glaucoma (or a family history of glaucoma), previous steroid myopathy or epilepsy. Liver failure.
- Corticosteroid effects may be enhanced in patients with hypothyroidism or cirrhosis and decreased in hyperthyroid patients.
- Diabetes may be aggravated, necessitating a higher insulin dosage. Latent diabetes mellitus may be precipitated.
- Menstrual irregularities may occur, and this possibility should be mentioned to female patients.
- Rare instances of anaphylactoid reactions have occurred in patients receiving corticosteroids, especially when a patient has a history of drug allergies.
- All corticosteroids increase calcium excretion
- Aspirin should be used cautiously in conjunction with corticosteroids in patients with hypoprothrombinaemia.
- This product contains 10mg/ml benzyl alcohol and must not be given to premature babies or neonates. Benzyl Alcohol may cause toxic reactions and anaphylactoid reactions in infants and children up to 3 years old.

#### Use in Children:

Sivkort injection is not recommended for children under six years. Corticosteroids cause dose-related growth retardation in infancy, childhood and adolescence which may be irreversible, therefore growth and development of children on prolonged corticosteroid therapy should be carefully observed.

#### Use in Elderly:

The common adverse effects of systemic corticosteroids may be associated with more serious consequences in old age, especially osteoporosis, hypertension, hypokalaemia, diabetes, susceptibility to infection and thinning of the skin. Close clinical supervision is required to avoid life-threatening reactions.

### **Interactions with Other Medicaments**

Amphotericin B injection and potassium-depleting agents: Patients should be observed for hypokalaemia.

- Anticholinesterases: Effects of anticholinesterase agent may be antagonised.
- Anticoagulants, oral: Corticosteroids may potentiate or decrease anticoagulant action. Patients receiving oral anticoagulants and corticosteroids should therefore be closely monitored.
- Antidiabetics: Corticosteroids may increase blood glucose; diabetic control should be monitored, especially when corticosteroids are initiated, discontinued, or changed in dosage.
- Antihypertensives, including diuretics: corticosteroids antagonise the effects of antihypertensives and diuretics. The hypokalaemic effect of diuretics, including acetazolamide, is enhanced.
- Anti-tubercular drugs: Isoniazid serum concentrations may be decreased.
- Cyclosporin: Monitor for evidence of increased toxicity of cyclosporin when the two are used concurrently.
- Digitalis glycosides: Co-administration may enhance the possibility of digitalis toxicity.
- Oestrogens, including oral contraceptives: Corticosteroid half-life and concentration may be increased and clearance decreased.
- Hepatic Enzyme Inducers (e.g. barbiturates, phenytoin, carbamazepine, rifampicin, primidone, aminoglutethimide): There may be increased metabolic clearance of Sivkort injection. Patients should be carefully observed for possible diminished effect of steroid, and the dosage should be adjusted accordingly.
- Human growth hormone: The growth-promoting effect may be inhibited.
- Ketoconazole: Corticosteroid clearance may be decreased, resulting in increased effects.
- Nondepolarising muscle relaxants: Corticosteroids may decrease or enhance the neuromuscular blocking action.
- Nonsteroidal anti-inflammatory agents (NSAIDs): Corticosteroids may increase the incidence and/or severity of GI bleeding and ulceration associated with NSAIDs. Also, corticosteroids can reduce serum salicylate levels and therefore decrease their effectiveness. Conversely, discontinuing corticosteroids during high-dose salicylate therapy may result in salicylate toxicity. Aspirin should be used cautiously in conjunction with corticosteroids in patients with hypoprothrombinaemia.
- Thyroid drugs: Metabolic clearance of adrenocorticoids is decreased in hypothyroid patients and increased in hyperthyroid patients. Changes in thyroid status of the patient may necessitate adjustment in adrenocorticoid dosage.
- Vaccines: Neurological complications and lack of antibody response may occur when patients taking corticosteroids are vaccinated.

### **Pregnancy and Lactation**

The ability of corticosteroids to cross the placenta varies between individual drugs, however triamcinolone does cross the placenta.

Administration of corticosteroids to pregnant animals can cause abnormalities of foetal development including cleft palate, intra-uterine growth retardation and effects on brain growth and development. There is no evidence that corticosteroids result in an increased incidence of congenital abnormalities, such as cleft palate / lip in man. However, when administered for prolonged periods or repeatedly during pregnancy, corticosteroids may increase the risk of intra-uterine growth retardation. Hypoadrenalism may, in theory, occur in the neonate following prenatal exposure to corticosteroids but usually resolves spontaneously following birth and is rarely clinically important.

As with all drugs, corticosteroids should only be prescribed when the benefits to the mother and child outweigh the risks. When corticosteroids are essential, however, patients with normal pregnancies may be treated as though they were in the non-gravid state.

#### Lactation

Corticosteroids may pass into breast milk, although no data are available for triamcinolone. Infants of mothers taking high doses of systemic corticosteroids for prolonged periods may have a degree of adrenal suppression.

#### Side Effects

Where adverse reactions occur they are usually reversible on cessation of therapy. The incidence of predictable side-effects, including hypothalamic-pituitary-adrenal suppression correlate with the relative potency of the drug, dosage, timing of administration and duration of treatment.

Absorption of triamcinolone following injection by the intra-articular route is rare. However, patients should be watched closely for the following adverse reactions which may be associated with any corticosteroid therapy:

Anti-inflammatory and immunosuppressive effects: Increased susceptibility and severity of infections with suppression of clinical symptoms and signs, opportunistic infections, recurrence of dormant tuberculosis.

Fluid and electrolyte disturbances: sodium retention, fluid retention, congestive heart failure in susceptible patients, potassium loss, cardiac arrhythmias or ECG changes due to potassium deficiency, hypokalaemic alkalosis, increased calcium excretion and hypertension.

Musculoskeletal: muscle weakness, fatigue, steroid myopathy, loss of muscle mass, osteoporosis, avascular osteonecrosis, vertebral compression fractures, delayed healing of fractures, aseptic necrosis of femoral and humeral heads, pathological fractures of long bones and spontaneous fractures, tendon rupture.

Hypersensitivity: Anaphylactic reactions, angiodema, rash, pruritus and urticaria, particularly where there is a history of drug allergies.

Dermatological: impaired wound healing, thin fragile skin, petechiae and ecchymoses, facial erythema, increased sweating, purpura, striae, hirsutism, acneiform eruptions, lupus erythematosus-like lesions and suppressed reactions to skin tests.

Gastrointestinal: dyspepsia, peptic ulcer with possible subsequent perforation and haemorrhage, pancreatitis, abdominal distension and ulcerative oesophagitis, candidiasis.

Neurological: euphoria, psychological dependence, depression, insomnia, convulsions, increased intracranial pressure with papilloedema (pseudo-tumour cerebri) usually after treatment, vertigo, headache, neuritis or paraesthesias and aggravation of preexisting psychiatric conditions and epilepsy.

Endocrine: menstrual irregularities and amenorrhoea; development of the Cushingoid state; suppression of growth in childhood and adolescence; secondary adrenocortical and pituitary unresponsiveness, particularly in times of stress (e.g. trauma, surgery or illness); decreased carbohydrate tolerance; manifestations of latent diabetes mellitus and increased requirements for

insulin or oral hypoglycaemic agents in diabetes, weight gain. Negative protein and calcium balance. Increased appetite.

Ophthalmic: posterior subcapsular cataracts, increased intraocular pressure, glaucoma, exophthalmos, papilloedema, corneal or scleral thinning, exacerbation of ophthalmic viral or fungal diseases.

Others: necrotising angitis, thrombophlebitis, thromboembolism, leucocytosis, insomnia and syncopal episodes.

Withdrawal Symptoms and Signs:

On withdrawal, fever, myalgia, arthralgia, rhinitis, conjunctivitis, painful itchy skin nodules and weight loss may occur. Too rapid a reduction in dose following prolonged treatment can lead to acute adrenal insufficiency, hypotension and death.

Intra-Articular Injection:

Reactions following intra-articular administration have been rare. In a few instances, transient flushing and dizziness have occurred. Local symptoms such as post-injection flare, transient pain, irritation, sterile abscesses, hyper- or hypo-pigmentation, Charcot-like arthropathy and occasional increase in joint discomfort may occur. Local fat atrophy may occur if the injection is not given into the joint space, but is temporary and disappears within a few weeks to months.

Intramuscular Injection:

Severe pain has been reported following intramuscular administration. Sterile abscesses, cutaneous and subcutaneous atrophy, hyperpigmentation, hypopigmentation and Charcot-like arthropathy have also occurred.

**Symptoms and Treatment of Overdose**

Since there is no special antidote, treatment of Triamcinolone Acetonide overdose or toxic reactions should be symptomatic and supportive.

**Presentation**

This injection is available in 1 ml ampoules  
10 ampoules are packed into one folded paper box.

**Storage Condition**

Protect from light and store below 30°C. Keep out of reach of children.

**Shelf Life**

The injections can be used within 48 months from the date of manufacturer if kept as recommended. Discard any unused portion.

**Registration number**

Sivkort sterile suspension for injection 40 mg-MAL08111829AZ

**Name and Address of Manufacturer**

SIU GUAN CHEM.IND.CO.,LTD  
No. 128 Shin Min Road, Chia Yi, Taiwan.  
Tel: 05-2360636  
Fax: 05-2223961

**Product Registration Holder**

Averroes Pharmaceuticals Sdn.Bhd  
3-08-01 & 3-09-01, Block 3  
Presint Alami, Worldwide Business Centre II  
Persiaran Akuatik, Section 13,  
40100 Shah Alam, Selangor  
Malaysia  
Tel: +603 55111433  
Fax: +603 55111431

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